Serendib University Endowment Fund — Portfolio Tear Sheet

USD 10m | Decision: ADJUST scenario limit to 18% | Sample: 19 Nov 2004–27 Oct 2026

Allocation and construction

ETF	Weight	Capital USD
SPY	9.55%	955,000
EFA	0.00%	0
EEM	0.00%	0
AGG	58.85%	5,885,000
GLD	15.00%	1,500,000
XLE	1.60%	160,000
XLK	15.00%	1,500,000

Full investment, original ETF bounds, 99% historical daily VaR ≤ 1.5%, ES ≤ 2.0%, and all five scenario losses ≤ 18% are enforced in construction. The 18% proposal was selected after reviewing results; it is not independently calibrated. If 15% is mandatory, reject this retained allocation under that mandate.

Performance and validation

Historical CAGR **7.42%**, volatility **7.16%**, Sharpe **0.786**, maximum drawdown **18.27%**. Return hurdles 7.0%/7.4%/7.8% are team assumptions, not forecasts. Weights were selected in sample. Walk-forward Maximum Sharpe: 7.10% return, Sharpe 0.807; Minimum variance: 6.82%, 0.839. Maximum Sharpe has no demonstrated walk-forward Sharpe advantage. Policy benchmark: 40% SPY/60% AGG; information ratio 0.197.

One-day risk on USD 10m

Model	VaR	ES
Historical simulation	1.163%	1.734%
Parametric (normal)	1.020%	1.173%
Parametric (Student t)	1.157%	1.726%
Monte Carlo	1.152%	1.674%
Joint ETF bootstrap	1.169%	1.735%
EVT / GPD	1.220%	1.726%

Historical 99% VaR USD 116,306; ES USD 173,429. Technology is 15% of money but about 36.5% of historical ES. EVT 99% bootstrap 95% intervals: VaR 1.123–1.299%, ES 1.488–1.966% (200 IID replicates; clustering may widen uncertainty).

Stress and decision

Scenario	Loss	Loss USD
2008 financial crisis	5.170%	517,024
March 2020 Covid crash	9.892%	989,191
2022 rate hikes	12.645%	1,264,453
Inflation shock	17.215%	1,721,500
Equity crash and dollar spike (assumption)	8.540%	854,000

Inflation loss is 17.215%, below revised 18% but above former 15%. The limit is scenario loss, **not drawdown**. A 15%-constrained alternative exists but worsens drawdown to 19.58%. Both hypothetical scenarios are assumptions, including XLE 0% in inflation. Energy rotation is not a universal hedge.

Monitoring and limitations

All compared models, including GARCH, still fail independence. GARCH filtered HS also fails coverage. Daily rebalancing/costless execution underlies headline results; quarterly sensitivity: 7.3% return, −18.7% drawdown. Liquidity reserves and multi-day risk are not implemented. Monitor Student-t, EVT, conditional diagnostics and scenarios; review on limit breaches, adverse correlations or a 99%/250-day Basel red signal. Scenario feasibility does not guarantee future compliance.